

Activation Choice Is a Design Variable in Informer Forecasting

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Abstract. Activation functions inside long-sequence forecasting transformers are usually inherited as implementation defaults, even though they control saturation, sign preservation, and gradient scale inside the feed-forward blocks. We test whether this default is harmless by running an activation-only Informer grid over ETTh1, ETTh2, ETTm1, ETTm2, and eight Solar sites: 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration, for 972 matched seed-runs. The architecture, attention type, width, depth, optimizer, data splits, batch size, and epoch budget are held fixed inside each cell. Non-GELU activations are best in 34 of 36 cells; only Solar3 at horizons 24 and 96 remains GELU-best. Bounded and tanh-family activations are the strongest static choices: `tanh_sin001`, `tanh`, and `softsign` have the best mean ranks and jointly account for 23 best cells. The same grid also rejects a universal replacement rule. Output `tanh` wins several high-gain ETT/ETTm cells but is poor on average, ReLU wins only one cell, and factor-oracle diagnostics provide weak leave-one-seed-out evidence for regime routing. PatchTST and iTransformer stress tests further show that the Informer ranking is not architecture-invariant. The supported conclusion is narrow but useful: under a matched Informer protocol, activation choice is a dataset- and horizon-sensitive design variable, not a harmless default.

Keywords: Time-series forecasting · Activation functions · Informer · Regime diagnostics

1 Introduction

Long-horizon time-series forecasting papers usually treat the model architecture as the primary design object. Attention sparsity, temporal decomposition, patching, channel mixing, and embedding design receive direct experimental attention, while the feed-forward activation is often inherited from a reference implementation. This convention is convenient, but it hides a real modeling choice. The activation function determines whether intermediate representations are sign-preserving or rectified, bounded or unbounded, smooth or piecewise linear, and whether large excursions are amplified or saturated. These properties are especially relevant in forecasting, where the same architecture is evaluated across horizons, frequencies, and datasets with different volatility and seasonal structure.

This paper asks a deliberately narrow question: if the Informer architecture and training protocol are held fixed, how much does the activation configuration matter? We choose Informer [7] as the controlled substrate because it is a canonical long-sequence forecasting transformer and its public scripts define horizon-specific settings for the ETT benchmarks. We do not propose a new attention mechanism, loss function, or

backbone. Instead, we expose the activation family and its placement as explicit experimental variables while keeping attention, embedding, depth, width, optimizer, data split, batch size, and epoch budget fixed inside each dataset-horizon cell.

The resulting answer is positive, but not in the form of a universal activation rule. In the completed Informer grid, non-GELU activations are best in 34 of 36 dataset-horizon cells. Bounded and tanh-family activations are the strongest static choices: `tanh_sin001_all`, `tanh_all`, and `softsign_all` have the best mean ranks and together win 23 cells. At the same time, the best configuration changes across datasets and horizons. Solar3 remains GELU-best at both tested horizons, output `tanh` wins several ETT/ETTh cells but is poor on average, and ReLU wins only one cell. Thus the main result is not that one activation replaces GELU everywhere. It is that activation choice is a first-order design variable under a fixed forecasting protocol.

Figure 1 summarizes this pattern. The large gains concentrate in ETTh2 and ETTm2, while Solar gains are smaller and site-specific. This heterogeneity motivates a second question: can input-window factors such as volatility, jump intensity, turbulence, or spectral concentration select the best activation without retraining? Our current evidence says no. Factor bins reveal regime dependence, but leave-one-seed-out oracle evidence is weak and localized. We therefore use factors as diagnostics for targeted follow-up studies, not as a deployed zero-shot router.

We also test whether the Informer ranking transfers to other forecasting architectures. PatchTST [4] shows partial transfer: non-GELU activations win selected ETTh2 and Solar cells, but GELU remains the best static choice in most tested cells, and output `tanh` loses every paired comparison against the matching linear head. iTransformer [2] is an even stricter stress test: in a one-seed FFN-only pilot, GELU remains best in five of six cells, while a later small placement probe suggests conditional output-tanh gains on selected cells. These results support an architecture-dependent story and prevent overclaiming direct transfer.

The contributions are:

1. We implement an activation-only Informer evaluation interface that changes hidden activation family and placement while holding the rest of the architecture and training protocol fixed.
2. We report a complete three-seed grid over 36 dataset-horizon cells, nine activation configurations, and 972 matched seed-runs, showing that non-GELU activations are best in 34/36 cells.
3. We identify bounded and tanh-family activations as strong static Informer choices while showing that no single activation or placement universally dominates.
4. We separate diagnostic regime evidence from deployment claims: factor bins expose conditional behavior, but current leave-one-seed-out results do not support a zero-shot activation router.
5. We include PatchTST and iTransformer stress tests to show that activation sensitivity is real but mediated by architecture and placement.

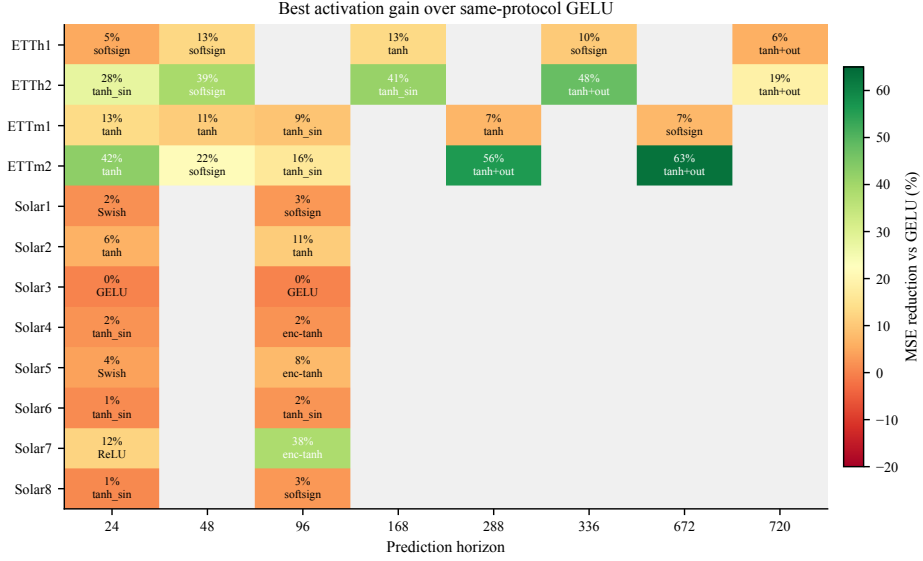


Fig. 1: Best activation gain over the same-protocol GELU row in each completed dataset-horizon cell. The full grid covers 36 dataset-horizon cells, nine activation configurations, and three seeds. Most cells prefer a non-GELU activation, but Solar3 remains GELU-best at both horizons.

2 Related Work

Transformer forecasting backbones. Transformer forecasting models adapt sequence attention to long input windows and multivariate horizons. Informer [7] uses ProbSparse attention and an encoder-decoder design to reduce the cost of long-sequence forecasting. PatchTST [4] changes the representation by treating a time series as a sequence of patches, while iTransformer [2] inverts the tokenization pattern so variables rather than time steps become the tokens. These works mainly change the architecture or representation. Our study is orthogonal: we keep the backbone fixed within each experiment and isolate the activation family and placement.

Activation functions as controlled design variables. ReLU [3] and GELU [1] are common neural-network defaults. GELU is particularly common in transformer implementations because it is smooth and has become part of many standard feed-forward blocks. In long-horizon forecasting code, however, the default activation is rarely treated as an experimental factor on the same footing as attention, width, patch length, or decomposition. Our results show why this is risky: under a matched Informer protocol, replacing the activation alone changes which configuration is best in nearly every dataset-horizon cell.

Bounded and periodic nonlinearities. Bounded nonlinearities such as `tanh` and `softsign` constrain activation magnitude while preserving sign. Periodic activations

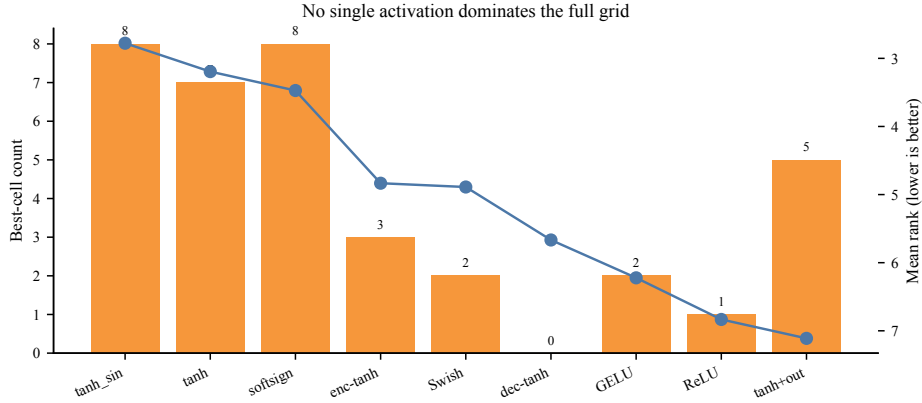


Fig. 2: Static activation ranking across the full Phase-4 grid. Bounded and tanh-family activations rank best on average, but the best-cell counts are spread across several configurations, arguing against a universal replacement activation.

and Fourier-style features have also been used to represent high-frequency functions and signals [5,6]. Our oscillatory `tanh_sin001` variant is much more modest than the periodic activations used in implicit neural representations: it adds a small sinusoidal perturbation around `tanh` inside a forecasting transformer’s feed-forward blocks. This makes the comparison against the unperturbed bounded baseline essential. The full grid shows that `tanh_sin001` is a strong static candidate, but it does not justify a general claim that oscillation is always better than `tanh`.

Regime-aware forecasting diagnostics. Forecasting errors depend on input-window regimes such as volatility, jumps, trend consistency, and seasonal concentration. It is therefore tempting to turn activation selection into a zero-shot routing problem: compute factors from the input window and choose an activation or placement. Our current evidence is more conservative. Factor bins reveal that activation differences vary across regimes, but retrospective bin contrasts are not deployment evidence. We therefore distinguish three levels of evidence: full-grid static performance, same-split diagnostic oracles, and leave-one-seed-out or held-out router validation. Only the first is currently strong enough to support the main paper claim.

Architecture transfer. If activation sensitivity were purely a property of the dataset, an activation ranking found in Informer should transfer directly to PatchTST or iTransformer. The transfer tests contradict that simple story. PatchTST shows selected non-GELU wins but keeps GELU as the strongest static choice in most tested cells. iTransformer shows even weaker FFN-only transfer in the current one-seed pilot. These negative and partial results are important because they define the scope of the contribution: activation choice matters, but it is jointly conditioned on dataset, horizon, placement, and architecture.

3 Activation Grid and Protocol

3.1 Forecasting setup

Given a multivariate input window $X_{t-L+1:t} \in \mathbb{R}^{L \times C}$, the model predicts the next H target values $Y_{t+1:t+H}$. We report mean squared error (MSE) as the primary metric and mean absolute error (MAE) as a secondary metric. For a configuration c , dataset d , horizon H , and seed s , let $m(c, d, H, s)$ denote the MSE. Improvements over GELU are reported as paired relative reductions:

$$\Delta(c, d, H, s) = \frac{m(\text{GELU}, d, H, s) - m(c, d, H, s)}{m(\text{GELU}, d, H, s)}. \quad (1)$$

Positive values mean that configuration c improves over the same-seed GELU run under the same dataset-horizon protocol.

The main protocol follows the official-script style used for Informer ETT experiments: ProbSparse attention, time-feature embedding, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, eight heads, dropout 0.05, learning rate 10^{-4} , patience 3, batch size 32, and six training epochs. We call this an official-script-style protocol rather than a reproduction of every original Informer number, because our central comparison is internal: activation-only changes against our own matched GELU baseline.

3.2 Activation families and placements

Let z be the pre-activation tensor in an Informer feed-forward block. The hidden activation families are

$$f(z) \in \{\text{GELU}(z), \text{ReLU}(z), \text{Swish}(z), \tanh(z), \text{softsign}(z), \tanh(z) + 0.01 \sin(z)\}. \quad (2)$$

GELU is the baseline. ReLU and Swish are non-bounded controls. $\tanh$, softsign , and $\tanh_{\text{sin001}}$ test whether bounded sign-preserving nonlinearities are useful in long-horizon forecasting.

Placement is controlled independently from the activation family. The main full-hidden configurations replace the hidden activation in both encoder and decoder feed-forward blocks. We also test encoder-only $\tanh$, decoder-only $\tanh$, and hidden $\tanh$ combined with an output $\tanh$. The output variant is treated separately because it changes the prediction range and can behave like target clipping rather than merely changing hidden representation.

3.3 Datasets and horizon cells

The Informer grid covers ETTh1, ETTh2, ETTm1, ETTm2, and eight Solar sites. ETTh cells use horizons 24, 48, 168, 336, 720; ETTm cells use 24, 48, 96, 288, 672; Solar cells use 24, 96. This gives 36 dataset-horizon cells. Each cell contains nine activation configurations and three seeds, yielding 324 dataset-horizon-configuration cells and 972 matched seed-runs.

The sequence length, label length, encoder depth, decoder depth, and ProbSparse factor are horizon-specific and are recorded in the experiment manifest. Within a dataset-horizon cell, these fields are fixed across all activation configurations. This design is intentionally narrower than a general hyperparameter search: it asks whether the activation choice matters when everything else is matched.

3.4 Aggregation and evidence policy

The main tables use only rows with batch size 32, six epochs, finite metrics, complete three-seed coverage, and matching protocol fields. Repeated remote rows are deduplicated by dataset, horizon, seed, protocol fields, and activation signature before aggregation. Rows from earlier small-batch Lee-style diagnostics, mismatched configs, incomplete cells, and non-finite runs are excluded from the main evidence.

For each dataset-horizon cell, configurations are ranked by mean MSE over the three seeds. Static rankings average these within-cell ranks over the 36 cells. Cell-level gain intervals are computed by first averaging paired-seed improvements inside each cell and then summarizing the 36 cell means. We use these intervals as descriptive uncertainty checks, not as a claim that the 36 benchmark cells are an i.i.d. sample from a larger population.

3.5 Regime diagnostics

To probe why different activations win in different cells, we compute input-window factors including volatility, realized volatility, volatility shock, jump intensity, max absolute change, tail ratio, trend consistency, reversal rate, mean-reversion strength, and spectral summaries. Summary-level factor bins cover the full completed grid. Sample-level oracles are available only for cells with saved prediction arrays.

We distinguish diagnostics from deployment. Same-split factor oracles are optimistic upper bounds because they observe the completed static runs. Leave-one-seed-out or held-out evaluations are stricter because they ask whether a factor rule generalizes beyond the seed used to estimate it. The current leave-one-seed-out evidence is too weak and localized to support a deployable activation router, so the paper treats regime factors as explanation and experiment-design tools.

4 Experiments

4.1 Full-grid coverage

The completed Informer matrix contains 972 eligible seed-runs: 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration. All 324 dataset-horizon-configuration cells have complete three-seed coverage under the matched batch-size-32, six-epoch protocol. Table 1 summarizes the coverage split across ETTh, ETTm, and Solar.

The aggregation is strict. A row is reusable only when dataset, horizon, sequence length, label length, layer counts, ProbSparse factor, batch size, epochs, activation signature, and seed all match the canonical key. The Solar4 source file initially produced

Table 1: Phase-4 full-grid coverage. A main-table cell is eligible only when all three seeds are present under batch size 32 and six training epochs.

Group	dataset-horizon	cells	config	cells	seed-runs	3-seed	config	cells
ETTh	10	90	270/270	90/90				
ETTh	10	90	270/270	90/90				
Solar	16	144	432/432	144/144				
All	36	324	972/972	324/324				

non-finite losses because six target values were missing; those values were repaired before relaunching the exact missing Solar4 seed-runs. The repaired source checksum is recorded in the artifact notes.

4.2 Activation-only changes beat GELU in most cells

Table 2 reports the best activation in each completed dataset-horizon cell. A non-GELU activation is best in 34/36 cells. The Wilson interval for this descriptive cell proportion is 81.9% to 98.5%, but the more important point is the breadth of the pattern: all 20 ETT/ETTh cells prefer a non-GELU activation, and 14 of 16 Solar cells do as well. The only GELU-best cases are Solar3 at horizons 24 and 96.

The largest best-cell gains occur in ETTh2 and ETTh2. ETTh2/336 and ETTh2/288 select hidden `tanh` with output `tanh`; ETTh2/672 selects the same output-bounded configuration and has the largest relative gain in the table. Solar gains are smaller and more site-specific. This is useful evidence against a trivial explanation that any non-GELU activation always helps: Solar3 keeps GELU, while Solar7 has high variance and a ReLU best cell at horizon 24.

4.3 Bounded and tanh-family activations are strong static choices

Table 3 ranks configurations within each dataset-horizon cell and averages the ranks over all 36 cells. The strongest static choices are `tanh_sin001_all`, `tanh_all`, and `softsign_all`. They also have high top-3 coverage and positive-cell rates against GELU. This supports the main positive lesson: bounded sign-preserving non-linearities are not merely one-off winners, but robust candidates under this Informer protocol.

Table 4 adds the uncertainty view used to keep this claim scoped. The top three static configurations have positive mean gains with descriptive intervals that remain above zero when summarized over dataset-horizon cells. In contrast, decoder-only `tanh` is weak, ReLU is negative on average, and output `tanh` has a strongly negative mean despite winning five cells. Therefore the paper should not recommend output bounding as a default. It should be treated as a targeted placement candidate for selected long-horizon ETT/ETTh cells.

4.4 Placement effects are real but conditional

The best-cell counts are spread across configurations. `tanh_sin001` and `softsign` each win eight cells, `tanh` wins seven, output `tanh` wins five, encoder-side `tanh` wins three, Swish wins two, GELU wins two, and ReLU wins one. Full-hidden bounded activations are the safest static candidates, but placement changes the result in important cells.

Output `tanh` is best in ETTh1/720, ETTh2/336, ETTh2/720, ETTm2/288, and ETTm2/672. These are high-gain cells, which explains why output bounding is tempting. However, the same placement has mean rank 7.11 and a negative average gain across the grid. Encoder-side `tanh` is best in Solar4/96, Solar5/96, and Solar7/96, while decoder-side `tanh` wins no cell. This pattern is consistent with placement being a conditional design variable rather than a universal trick.

4.5 Regime factors are diagnostic, not a router

Summary-level factor bins cover all 36 dataset-horizon cells and produce 783 factor-bin rows. They reveal substantial regime dependence: 196 factor-conditioned contrasts have absolute end-minus-start improvement at least 0.02. The strongest contrasts are often warnings rather than positive rules. For example, high volatility, jump intensity, and turbulence bins frequently show that output `tanh` becomes worse, not that a simple high-volatility rule should select it.

Sample-level leave-one-seed-out oracles are stricter. Among 84 leave-one-seed-out oracle rows, only seven are positive, all on Solar1/96. Table 5 lists these cases. The gains are small relative to the static grid and are not broad enough to justify deployment. The current conclusion is therefore negative for a zero-shot activation router: factors are useful for explaining conditional behavior and selecting follow-up ablations, but not for selecting a concrete activation and placement in a new cell.

4.6 Architecture-transfer stress tests

Table 6 and Table 7 test whether the Informer findings transfer to PatchTST. The answer is partial. In the FFN-linear stage, GELU remains best in 7 of 12 cells, while `softsign` and `tanh_sin001` win selected ETTh2 and Solar cells. This shows that activation sensitivity does not disappear, but the Informer static ranking does not transfer directly. The output-placement result is stronger and negative: output `tanh` loses all 54 paired seed-run comparisons against the matching linear-head configuration.

Table 8 gives a second transfer stress test. In a one-seed ETTh2/ETTh2 FFN-only pilot, GELU remains best in five of six cells, and the only non-GELU best case is a narrow `tanh_sin001` gain on ETTm2/96. A later three-cell placement probe gives a more conditional signal: output `tanh` helps on ETTh2/96 and ETTm2/720 but hurts on ETTm2/96. Because this placement probe is currently single-seed, we use it only as a direction for follow-up, not as a main statistical claim.

Together, the transfer experiments sharpen the story. The Informer full-grid claim is strong because it is complete and matched. The broader cross-architecture claim is weaker but still useful: activation behavior depends on architecture and placement, so an activation policy should not be promoted as a universal dataset-only rule.

Table 2: Best activation per dataset-horizon in the completed Phase-4 grid. Improvements are relative MSE reductions against the same-protocol GELU row with matched seeds; GELU-best cells have no relative-improvement entry.

Dataset	H	best config	MSE	gain vs GELU	seed win rate
ETTh1	24	softsign	0.532 ± 0.026	+5.2%	66.7%
ETTh1	48	softsign	0.646 ± 0.073	+12.6%	100.0%
ETTh1	168	tanh	0.912 ± 0.131	+12.9%	66.7%
ETTh1	336	softsign	1.219 ± 0.042	+10.4%	100.0%
ETTh1	720	tanh+out	1.328 ± 0.030	+6.0%	100.0%
ETTh2	24	tanh_sin	0.516 ± 0.023	+27.8%	100.0%
ETTh2	48	softsign	1.606 ± 0.386	+38.8%	100.0%
ETTh2	168	tanh_sin	2.963 ± 0.509	+41.3%	100.0%
ETTh2	336	tanh+out	2.054 ± 0.219	+47.7%	100.0%
ETTh2	720	tanh+out	3.298 ± 0.186	+18.6%	100.0%
ETTM1	24	tanh	0.314 ± 0.015	+13.4%	100.0%
ETTM1	48	tanh	0.453 ± 0.023	+10.8%	100.0%
ETTM1	96	tanh_sin	0.559 ± 0.050	+9.3%	100.0%
ETTM1	288	tanh	0.802 ± 0.044	+7.2%	66.7%
ETTM1	672	softsign	0.922 ± 0.023	+7.3%	66.7%
ETTM2	24	tanh	0.685 ± 0.191	+42.5%	100.0%
ETTM2	48	softsign	0.309 ± 0.014	+21.6%	100.0%
ETTM2	96	tanh_sin	1.335 ± 0.214	+16.4%	100.0%
ETTM2	288	tanh+out	2.355 ± 0.359	+55.7%	100.0%
ETTM2	672	tanh+out	2.851 ± 0.162	+62.7%	100.0%
Solar1	24	Swish	0.150 ± 0.000	+1.7%	66.7%
Solar1	96	softsign	0.224 ± 0.001	+2.6%	100.0%
Solar2	24	tanh	0.066 ± 0.003	+6.4%	100.0%
Solar2	96	tanh	0.108 ± 0.005	+10.8%	100.0%
Solar3	24	GELU	0.198 ± 0.009	—	—
Solar3	96	GELU	0.349 ± 0.007	—	—
Solar4	24	tanh_sin	0.220 ± 0.003	+1.9%	100.0%
Solar4	96	enc-tanh	0.290 ± 0.005	+1.9%	66.7%
Solar5	24	Swish	0.109 ± 0.002	+4.2%	100.0%
Solar5	96	enc-tanh	0.180 ± 0.003	+7.6%	100.0%
Solar6	24	tanh_sin	0.265 ± 0.002	+1.1%	100.0%
Solar6	96	tanh_sin	0.358 ± 0.009	+2.3%	100.0%
Solar7	24	ReLU	3.281 ± 1.035	+12.0%	66.7%
Solar7	96	enc-tanh	4.970 ± 1.353	+38.2%	100.0%
Solar8	24	tanh_sin	0.216 ± 0.003	+0.8%	100.0%
Solar8	96	softsign	0.306 ± 0.005	+2.6%	100.0%

Table 3: Static activation ranking across the 36 completed dataset-horizon cells. Rank is computed within each cell over the nine Phase-4 configurations.

Config	mean rank	best cells	top-3 cells	mean gain	positive cells
tanh_sin	2.78	8	27	+11.7%	83.3%
tanh	3.19	7	25	+11.9%	80.6%
softsign	3.47	8	22	+11.2%	80.6%
enc-tanh	4.83	3	11	+5.1%	66.7%
Swish	4.89	2	5	+3.6%	77.8%
dec-tanh	5.67	0	3	+1.9%	66.7%
GELU	6.22	2	4	+0.0%	0.0%
ReLU	6.83	1	4	-2.3%	36.1%
tanh+out	7.11	5	7	-36.3%	25.0%

Table 4: Cell-level uncertainty for static Informer activation choices. The mean gain is computed from paired-seed relative MSE improvements over same-protocol GELU, first averaged inside each dataset-horizon cell and then summarized over the 36 cells. Intervals are descriptive 95% normal intervals over cells, not a claim of population-level optimality.

Config	mean gain with 95% interval	positive cells	best cells	top-3 cells
tanh_all	+11.9% [+6.6, +17.3]	29/36	7	25
tanh_sin001_all	+11.7% [+5.7, +17.7]	30/36	8	27
softsign_all	+11.2% [+5.3, +17.0]	29/36	8	22
enc_tanh_dec_gelu	+5.1% [+1.3, +9.0]	24/36	3	11
swish_all	+3.6% [+0.5, +6.8]	28/36	2	5
enc_gelu_dec_tanh	+1.9% [-0.7, +4.5]	24/36	0	3
relu_all	-2.3% [-4.6, 0.0]	13/36	1	4
tanh_all_outtanh	-36.3% [-58.8, -13.9]	9/36	5	7

Table 5: Top positive leave-one-seed-out factor-oracle diagnostics among array-ready Phase-4 cells. These are diagnostic upper bounds for regime sensitivity, not deployed activation-routing results.

Dataset	H	factor	LOSO oracle MSE	best static MSE	gain
Solar1	96	max_abs_change	0.174	0.178	+2.0%
Solar1	96	volatility	0.175	0.178	+1.4%
Solar1	96	volatility_shock	0.175	0.178	+1.3%
Solar1	96	trend_consistency	0.177	0.178	+0.4%
Solar1	96	max_drawdown	0.177	0.178	+0.2%
Solar1	96	max_drawup	0.177	0.178	+0.2%
Solar1	96	range	0.177	0.178	+0.2%

Table 6: PatchTST activation-transfer stress test. The Informer activation effect transfers only partially: non-GELU activations win selected ETTh2/Solar cells, while GELU remains best in 7 of 12 tested cells.

Model	Dataset	Horizon	GELU MSE	Best config	Gain
PatchTST	ETTh2	168	0.325880	gelu_ffn_linear	0.00%
PatchTST	ETTh2	336	0.328450	gelu_ffn_linear	0.00%
PatchTST	ETTh2	720	0.380881	softsign_ffn_linear	0.67%
PatchTST	ETTh2	96	0.165589	gelu_ffn_linear	0.00%
PatchTST	ETTh2	288	0.262739	gelu_ffn_linear	0.00%
PatchTST	ETTh2	672	0.355232	gelu_ffn_linear	0.00%
PatchTST	Solar2	24	0.065450	tanh_sin001_ffn_linear	0.42%
PatchTST	Solar2	96	0.099666	softsign_ffn_linear	1.12%
PatchTST	Solar3	24	0.279035	gelu_ffn_linear	0.00%
PatchTST	Solar3	96	0.446409	gelu_ffn_linear	0.00%
PatchTST	Solar5	24	0.114792	tanh_sin001_ffn_linear	1.33%
PatchTST	Solar5	96	0.176366	softsign_ffn_linear	1.01%

Table 7: PatchTST output-tanh placement test. Output tanh loses every paired seed-run comparison against the matching linear-head configuration, indicating that the Informer output-bounding signal does not transfer to PatchTST.

Base activation	Paired seed-runs	Win rate	Mean gain vs. linear
tanh	18	0.0%	-5.28%
tanh_sin001	18	0.0%	-5.31%
gelu	18	0.0%	-5.44%

Table 8: iTransformer activation-transfer stress test. In this one-seed pilot, GELU remains best in 5 of 6 ETTh2/ETTm2 cells; the only non-GELU win is a narrow `tanh_sin001` gain on ETTm2/96.

Model	Dataset	Horizon	GELU MSE	Best config	Gain
iTransformer	ETTh2	96	0.301598	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTh2	336	0.422894	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTh2	720	0.428986	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTm2	96	0.184226	<code>tanh_sin001_ffn_linear</code>	0.19%
iTransformer	ETTm2	336	0.313418	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTm2	720	0.414092	<code>gelu_ffn_linear</code>	0.00%

5 Conclusion

This paper isolates activation choice as a controlled design variable in Informer forecasting. Under a matched official-script-style protocol, activation-only changes alter the best configuration in most dataset-horizon cells: non-GELU activations are best in 34 of 36 completed cells, and bounded or tanh-family configurations have the strongest static ranks. The result is practically useful because it does not require changing attention, width, depth, or the training schedule.

The evidence also sets clear limits. `tanh_sin001`, `tanh`, and `softsign` are strong Informer candidates, but none is universal. Output `tanh` has several high-gain wins and a poor average rank. Decoder-only `tanh` is weak. Solar gains are smaller and site-specific, with Solar3 remaining GELU-best at both tested horizons. Factor diagnostics reveal regime dependence, but current leave-one-seed-out evidence does not support a deployable zero-shot activation router.

The strongest current design lesson is therefore conservative: GELU should not be treated as a harmless default in Informer-style forecasting, but activation choice should be validated per dataset, horizon, placement, and architecture. Future work should turn the diagnostic factor evidence into prospective routing experiments, expand the iTransformer placement probe to more seeds and cells, and test whether architecture-aware rules can select candidate activations before training.

References

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A Artifact and Reproducibility Notes

The completed Informer matrix is stored under the repository’s Phase-4 analysis directory. The aggregate files used by the paper are:

- phase4_protocol_raw_dedup.csv
- phase4_protocol_aggregate.csv
- phase4_completion_matrix.csv
- phase4_paired_vs_gelu.csv
- phase4_factor_bin_summary.csv
- phase4_factor_contrast_summary.csv
- phase4_factor_oracle_summary.csv
- phase4_factor_oracle_loso_summary.csv

Tables and figures are generated by the Phase-4 paper-asset script. The activation code lives in the model activation module, with encoder and decoder injection handled in the corresponding model modules. Remote execution scripts are stored under the repository’s experiment script directory.

B Dataset Integrity

The Phase-4 preflight required all 12 local CSV files to exist with the expected schema and target column. Solar4 initially produced non-finite losses. The root cause was six missing values in `Solar/Site_4_130MW.csv`; these were filled using a minimal local policy and the exact Solar4 missing seed-runs were relaunched. The repair audit stores the file checksum and relaunch details.

C Excluded Diagnostics

Batch-adjusted Lee probes and earlier small-batch diagnostics are not part of the Phase-4 main evidence. They are retained in the repository for hypothesis generation, but every claim in the main paper is based on the matched batch-size-32, six-epoch, three-seed matrix.

D Current Evidence Limitations

The main limitation is that our GELU baseline does not fully reproduce the original Informer Table 2 values. This prevents a claim of reproducing and surpassing the original paper. The present claim is internal and controlled: under the same matched protocol, activation-only replacements often improve over GELU. A second limitation is that factor-bin results are diagnostic rather than causal. They motivate targeted ablations but do not establish a universal volatility rule or a deployable activation router.

E Use of AI Tools

The authors used LLM-based writing tools for drafting support, organization, and copy editing. The experimental design, code execution, result selection, numerical verification, scientific interpretation, and final manuscript responsibility remain with the human authors.